Hashim Khan

Senior Mechanical Design Engineer
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Open to relocation to the UK and EU | Requires visa sponsorship

PROFESSIONAL SUMMARY

Senior Mechanical Design Engineer with 9 years of experience taking electromechanical machines from concept to series production across circular-economy, robotics, industrial automation and precision engineering. Led mechanical design of Reverse Vending Machines and compactors for a state-wide Deposit Return System, scaling production from 12 to 18 machines per day. This is directly relevant to the deposit return and packaging-recycling schemes now growing across the UK and EU. Strong in design for manufacture and assembly, GD&T, tolerance stack-up, FEA and change control, working to CE marking, Machinery Directive 2006/42/EC (transition to Regulation (EU) 2023/1230) and ISO safety standards. Works closely with electrical, software, quality and supplier teams to deliver compliant, cost-effective, maintainable products. Seeking a sponsored Mechanical Design Engineer role in the UK or EU.

CORE COMPETENCIES

Mechanical Design & CAD (SolidWorks, Siemens NX,CATIA V5) | GD&T (ASME Y14.5) & Tolerance Stack-Up | Engineering Calculations & FEA | DFM/DFA | Multilevel BOM Creation & Engineering Change Management | Sheet Metal, CNC Machining, 3D Printing & Injection Moulding | Prototyping & Validation (EVT/DVT) | Mechatronic Integration (Sensors, Actuators, PCBs) | Vendor Development & Management | Cross-Functional Collaboration | Mentoring | Technical Documentation | Assembly Optimization | Mass optimization

PROFESSIONAL EXPERIENCE

Senior Mechanical Design Engineer | Retearn Technologies Pvt. Ltd. Dec 2025 – Present
Hyderabad, India

- Design and develop mechanical hardware and enclosures within a shared 3D CAD assembly, managing configuration, multilevel BOMs and PLM data (3DEXPERIENCE) so models, drawings and interfaces stay consistent as changes are released, design follows CE, ISO and EU Machinery Directive (2006/42/EC) principles.
- Led R&D, design, development and testing of a compactor machine, achieving 80% volume reduction and significantly improving storage-volume utilisation and collection efficiency.
- Engineered sheet metal enclosures, FRP composite discs and brackets, covering material selection, tolerance stack-up and K-factor optimisation. Designed high-speed knife-edge conveyors and inline singulation systems for smooth material flow.
- Developed material segregation mechanisms and drive systems integrated with electromechanical systems like IoT sensors, motors, cable routing, and AI-based computer vision for automated detection and sorting, electronics integration, cable routing, sensor mountings and heat flow management.
- Reduced part count and manufacturing cost, led root cause analysis (RCA) and assembly improvements, and managed BOM/ECN processes and vendor development, raising production from 12 to 18 machines per day (50% increase).
- Planned and executed FAT/SAT validation for performance and quality compliance. Mentored junior engineers and worked with electrical, embedded software and manufacturing teams.

Senior Mechanical Engineer | OEDEC Engineering Pvt Ltd Jan 2025 – Nov 2025
Pune, India

- Developed mechanical systems for marine robotics (ROV) and Special Purpose Machines (SPMs) in SolidWorks (3DEXPERIENCE), applying tolerance stack-up and FEA (structural, thermal, vibration) to design precision transmissions and sealed enclosures for harsh operating environments.
- Selected materials, heat treatments and advanced manufacturing processes (CNC machining, welding, additive manufacturing) to achieve tight tolerances in line with ISO/CE/DIN standards.
- Applied DFM/DFA and continuous-improvement methodologies (Lean, Kaizen, 5S) while managing CAD configuration, BOMs, GD&T and tolerance-stack documentation, reducing design-to-manufacture lead time by 15%, and developed a vision-based fastener counting system for automotive shop-floor operations.
- Collaborated with quality assurance and suppliers on CMM, NDT and compliance inspections, supporting validation and verification documentation to European regulatory requirements.

Assistant Manager | Kalyani Group, Bharat Forge Ltd Jun 2023 – Jan 2025
Pune, India

- Led design, development, testing and validation of defence-grade barrels, breech mechanisms and precision machine tools for major artillery programmes (ATAGS 155×52, MGS 155×39, Garuda 105, SRGM 76), extensively applying GD&T, material selection and tolerance stack-up analysis.
- Designed 3D CAD models, manufacturing drawings, long machining test fixtures, SPM's and inspection test rigs in Siemens NX and Solid Edge, developing MQP, ITP and tooling SOPs to ISO/DEF-STAN standards.

- Performed structural, thermo-mechanical and vibration FEA alongside ultrasonic testing, magnetic particle inspection and optical borescope visual testing to validate performance and reliability.
- Led cross-functional teams on machining operations, applying Lean, 5S, Kaizen and TPM to cut man-hours by 12% and raise production output by 40%, achieving in-house manufacturing capability through new tooling.
- Managed engineering documentation (BOMs, ECNs, inspection reports), analysed production cycle-time data for forecasting, and planned tooling CAPEX for shop-floor budgets.

Engineer R&D | Kalyani Group, Kalyani Strategic Management Services TSPL Aug 2020 – Jun 2023

Pune, India

- Led product development of vibration sensor enclosures and DAQ systems through multiple design iterations, conducting material R&D across wood, aluminium, plastic and glass to balance strength, durability and aesthetics.
- Engineered Sanjeevani, a COVID-19 SpO2-based Smart Oxygen Delivery System — designing the sensor, 3D-printed medical-grade enclosure and electro-pneumatic oxygen control unit under significant time pressure.
- Designed Special Purpose Machines (SPMs) for computer-vision-based inspection-UV-A/Visible spectrum and crack detection, and developed electromechanical pick-and-place robotic cells for drive shaft and crankshaft handling.
- Sensorized legacy machines for IIoT digital connectivity, building sensor data pipelines feeding AI/ML predictive-maintenance models, and maintaining CAD data, BOMs and validation reports to CE/ISO standards.

Mechanical Designer | Beauto Systems Jan 2020 – Aug 2020

Pune, India

- Led R&D for the company's first robotics project, developing AGV and robotic arm algorithms.
- Designed and built robotic automation systems for food processing, applying DFM/DFA principles to ISO/CE standards and producing GD&T-based fabrication drawings in SolidWorks.

Mechanical Product Developer | Footloose Labs Apr 2019 – Aug 2019

Bengaluru, India

- Developed 5–6 DOF robotic arms with integrated drive systems for pharmacy warehouse automation, performing engineering calculations and simulations in MATLAB, Octave, SolidWorks and ANSYS.
- Coordinated vendor relationships, inventory management and prototype fabrication, running mechanical testings.

Design Engineer | Aliyance Mechatronics Aug 2017 – Mar 2019

Pune, India

- Contributed to Roboseum, Gujarat Science City's robotics museum project with the Government of Gujarat, designing and simulating robotic systems including arm manipulators, rotary platforms and XYZ gantries.
- Designed and analysed robotic parts, SPM's and assemblies in SolidWorks to ISO/CE standards, producing prototypes via 3D printing, and led technical recruitment from leading engineering institutes.

KEY PROJECTS

- 2026 | Reverse Vending & Material Handling Systems: RVMs, conveyor systems, compactors, bottle sorting and singulation mechanisms, Bulk RVM's.
- 2025 | Marine ROV & Automation Systems: Remotely Operated Vehicles (ROVs) for cargo ship hull inspection and cleaning, special-purpose machinery (SPM) and automation systems.
- 2021–2025 | Industry 4.0, Robotics SPM & Defence Engineering: IIoT integration, shop-floor automation and digitalisation, robotics, special-purpose machinery, and defence artillery programmes.

EDUCATION

Bachelor of Technology (B.Tech.), **Mechanical Engineering** | Dr. APJ Abdul Kalam Technical University | 2013 – 2017

CERTIFICATIONS

Certified SolidWorks Professional (CSWP) | SOLIDWORKS 2024 Advanced Part Modeling | Geometric Dimensioning & Tolerancing (GD&T) | Siemens NX & CATIA V5 (Automotive, Industrial)

TECHNICAL SKILLS

- CAD/CAE: Siemens NX, SolidWorks (CSWP), Solid Edge, Creo, AutoCAD, ANSYS, engineering calculations, FEA/simulation
- Manufacturing: Sheet metal fabrication, CNC machining, 3D printing, injection moulding, die casting, tooling & fixtures
- Engineering: GD&T (ASME Y14.5), tolerance stack-up, DFM/DFA, multilevel BOM creation, engineering change management (ECN), risk assessment
- Data & Process: CAD revision control, structured document/PDM, NPI/product industrialisation, EVT/DVT validation
- Standards & Quality: ASME Y14.5, ISO 9001, ISO 10218, ISO 1101, ISO 2768, BS 8888, CE, DEF-STAN
- Process & Tools: SAP, Lean, 5S, Kaizen, TPM, root cause analysis, supplier development, Microsoft Office, basic Python
- Languages: English (professional working proficiency), Hindi (Native), German (*currently learning beginner level A1*)